EV Charging Suitability Analysis in Tampa, Florida

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Introduction

The growing demand for electric vehicles (EVs) has increased the need for charging infrastructure in urban regions. Effective planning for new charging stations requires evaluating both the physical environment and the spatial distribution of human activity. This project focuses on identifying the most suitable locations for future EV charging stations within the City of Tampa using a GIS-based multi-criteria suitability model. Our goal was to integrate environmental constraints, accessibility factors, and population characteristics to highlight areas that best support EV infrastructure growth.

To accomplish this, we used spatial analysis tools in ArcGIS Pro to combine several key variables: proximity to major roadways, population density, land cover compatibility, and environmental constraints such as flood zones and wetlands. By reclassifying each dataset onto a common suitability scale and applying a weighted overlay, we generated a final suitability map that highlights optimal zones for EV station placement. The results of this study can support planning decisions, improve accessibility for EV users, and promote more sustainable transportation infrastructure across Tampa.

Study Area

The study area for this project is the City of Tampa, located in Hillsborough County on the west coast of Florida. Tampa is one of the largest and fastest-growing metropolitan regions in the state, characterized by a diverse mix of residential neighborhoods, commercial districts, major transportation corridors, and environmentally sensitive areas. The city offers a representative landscape for evaluating electric vehicle (EV) infrastructure planning because it contains both high-demand urban centers and large expanses of wetlands and flood-prone land. Tampa's rapid population growth and increasing adoption of electric vehicles make it a relevant and important location for developing spatial models that support EV station expansion.

Tampa's geography also presents unique planning challenges. The region contains several major highways—including Interstate 275, Interstate 4, and the Selmon Expressway—that serve as primary transportation routes for daily commuters. These corridors are of particular interest in suitability modeling because EV charging stations

located near high-accessibility areas tend to be more heavily used and more economically viable. At the same time, Tampa is surrounded by extensive coastal and inland wetland systems and includes large areas designated by FEMA as Special Flood Hazard Areas. These environmental constraints must be incorporated into any infrastructure planning model in order to avoid constructing charging stations in areas with high flood risk or ecological sensitivity.

The spatial extent used for all analyses was defined by the Tampa city boundary, which was also used to clip all other datasets to create a consistent processing area. A uniform projected coordinate system (Florida State Plane West, NAD 1983, US Feet) was applied to all layers to ensure accurate distance measurements throughout the GIS workflow. Since the suitability model relies on raster-based overlay techniques, several datasets originally stored as vector layers were converted to raster format to maintain compatibility across all tools used in ModelBuilder.

Data

A variety of datasets were incorporated into this project in order to represent accessibility, human activity patterns, environmental constraints, and the physical landscape. Each dataset played a unique role in the suitability analysis. The road network was used to measure distance from major transportation corridors, which is a widely used indicator of EV charging accessibility. Areas closer to main roads are more convenient for users and are often preferred by planners. Likewise, population density was included to represent areas of greater potential charging demand. Regions with higher population concentrations are more likely to support frequent EV usage and benefit from additional charging stations.

Land cover was incorporated to help determine where new infrastructure would be realistically and appropriately located. Urban and developed land uses generally support the installation of EV charging stations, while wetlands, water bodies, and other ecologically sensitive areas are unsuitable and may require exclusion from the model. Additional constraint layers, including FEMA flood zones and National Wetlands Inventory polygons, were used to create binary constraint masks that removed areas

considered unsafe or environmentally restricted. These masks ensured that even if a location scored highly on human activity or accessibility factors, it would not be considered suitable if it overlapped with high-risk or environmentally sensitive terrain.

Each dataset underwent preprocessing steps including clipping, projecting, converting to raster format, reclassifying values into standardized suitability categories, and applying binary constraints where appropriate. Table 1 summarizes the datasets used in the project, their sources, and the manipulations performed prior to their inclusion in the suitability model. These transformations were essential to prepare the data for weighted overlay analysis, which requires all inputs to share the same resolution, extent, and standardized suitability scale.

The combination of data sources selected for this study provides a comprehensive view of both human and environmental considerations that influence EV charging station siting. Together, these layers allow the model to highlight locations that balance accessibility, demand, environmental safety, and land-use practicality. By integrating these datasets within a GIS-based framework, the project supports informed decision-making for EV infrastructure planning in the City of Tampa.

Dataset	Description	Source	Data Manipulation
Road Network	Street centerlines and major transportation routes used to calculate Euclidean distance for accessibility.	Florida Department of Transportation (FDOT) / project dataset	Projected to common coordinate system, clipped to Tampa boundary, converted to Euclidean Distance raster, and reclassified onto 1–5 suitability scale.
Population Density (ACS)	Census population counts and density estimates	U.S. Census Bureau, American Community Survey	Joined to geographic units, normalized to

	representing potential charging demand.	(2023)	density values, projected, converted to raster, and reclassified into five demand-based classes.
Land Cover	Land-use categories representing suitability for construction and development (urban areas, vegetation, wetlands, water, etc.).	Southwest Florida Water Management District (SWFWMD) / local government land-cover data	Clipped to the study area, cleaned, converted to raster, and reclassified into suitability classes where developed areas received higher scores and wetlands/water received low scores.
FEMA Flood Zones	Flood-prone areas used as environmental constraints to prohibit placement in unsafe locations.	FEMA National Flood Hazard Layer	Clipped to Tampa boundary, converted to raster, and reclassified into a binary mask (1 = safe X zones, 0 = Special Flood Hazard Areas).
Wetlands	Ecologically sensitive wetlands throughout Tampa Bay that limit where new infrastructure can	National Wetlands Inventory / SWFWMD	Clipped, converted from polygons to raster, and used to create a binary mask where wetland cells were

	be constructed.		assigned 0 (unsuitable) and all other cells were assigned 1 (suitable).
EV Charging Stations	Locations of existing EV charging stations used for reference and validation of the suitability model.	Provided project dataset	Projected and clipped to the study area and overlaid with the final suitability map to evaluate coverage and identify gaps.
Tampa Boundary	Administrative boundary defining the study area extent for all analyses.	City of Tampa Open Data	Used to clip every dataset, set the processing extent, and act as a mask in raster operations.

Methods

The approach that we took was to integrate different spatial datasets which we used to represent accessibility, demand, environmental, and physical constraints. Before, the final overlay any initial vector datasets were converted to a consistent raster format by using a defined cell size. The continuous layers that represented accessibility and demand were the layers like the Road Network distance and the Population Density which were standardized by reclassifying the values into a common suitability scale. The suitability scale went from 1 to 5, where 1 represented the lowest suitability and 5 represented the highest suitability. For the Road Networks any areas that were near the high, such as I-275 and I-4, they recieved higher suitability scores because these areas receive a lot more traffic and are frequently used which would make any new EV charging station implemented be highly used. As for the Population Density, any regions with higher

population counts were reclassified, and shown to have higher suitability scores, which indicated that the areas had a greater potential charging demand.

Two environmental constraints that were modeled as binary masks to enforce strict limitations on placement were FEMA flood zones and ecologically sensitive wetland areas from the National Wetlands Inventory. The FEMA flood zones were used to see which areas of Tampa were flood prone and with that information a binary constraint was applied to create a binary constraint mask. Any location that fell within a "Special Flood Hazard" area was automatically assigned a value of zero to permanently prohibit any possible placement of an EV charging station. With the National Wetlands Inventory any ecologically sensitive wetlands area were also assigned a value of zero. This was to avoid any destruction of local wetlands since they removed pollutants, reduce erosion, and are vital habitats for many plants and animals. Wetland soils are also very unstable which could increase any construction cost and would like the EV charging stations be very vulnerable to damage.

All spatial analysis was performed in ArcGIS Pro using ModelBuilder so that the workflow could be documented and repeated. After clipping and projecting the input layers, vector datasets such as roads, land cover, wetlands, and flood-safe zones were converted to raster format at a common cell size. Euclidean Distance was then run on the major road network to create a continuous distance surface, which was reclassified so that cells closer to highways and major arterials received higher suitability values. Population density rasters were grouped into five equal intervals that represented very low to very high potential demand, and these classes were assigned scores from one to five. Land-cover rasters were simplified so that urban and built-up categories received the highest suitability scores, agricultural and open areas received moderate scores, and wetlands or water were given the lowest scores.

Environmental constraints were incorporated through two separate binary masks representing FEMA flood zones and wetlands. Only X flood zones, which represent relatively low flood risk, were allowed to remain suitable in the model. All Special Flood Hazard Areas were given a value of zero so that they would be permanently excluded from possible EV locations. Wetland polygons from the National Wetlands Inventory

were treated similarly; once converted to raster, wetland cells were assigned a value of zero to avoid placing infrastructure in ecologically sensitive or physically unstable areas. These two masks were multiplied together to create a single environmental constraint grid that removed any cell that overlapped with either a flood zone or wetland.

Once each factor had been reclassified to a 1–5 suitability scale and the constraint mask was prepared, a weighted overlay was performed to combine the criteria. Proximity to major roads was given a weight of 40 percent because accessibility is a key requirement for successful charging stations. Land-cover suitability was weighted at 35 percent to emphasize the importance of locating stations on appropriate developed land, and population density was weighted at 25 percent to represent demand without overwhelming the physical constraints. The weighted overlay produced a composite suitability surface with values ranging from low to high. Finally, the highest-ranking cells were extracted and used to generate point features that represent candidate locations for new EV charging stations. These new points were then compared visually with the existing EV station layer to evaluate where coverage is currently lacking.

Results and Discussion

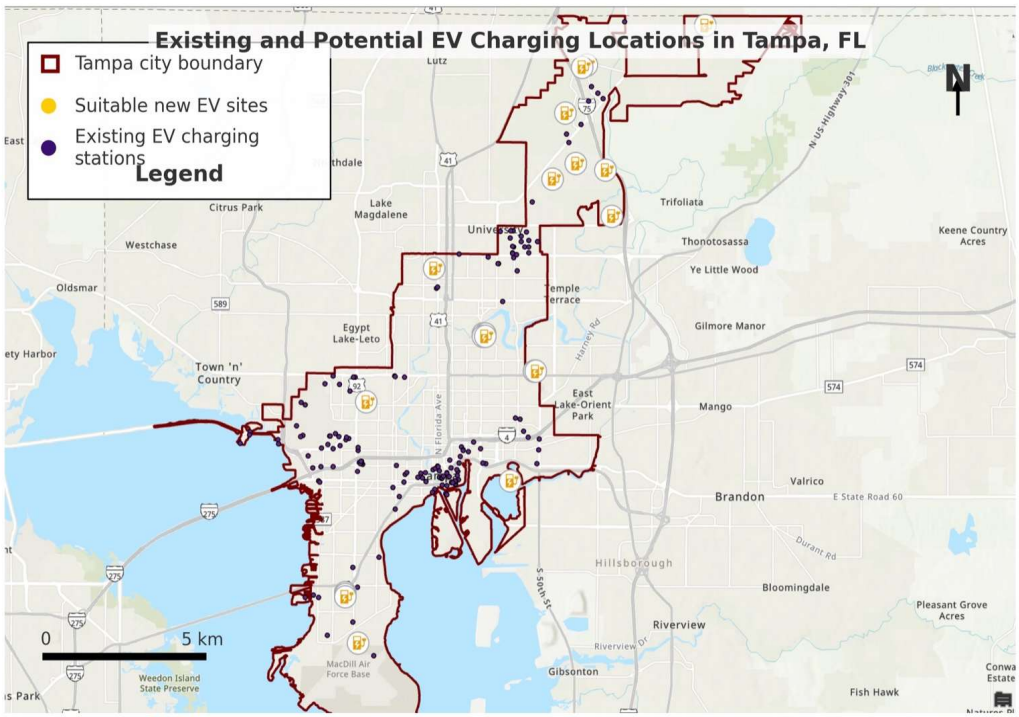
The final suitability surface identified several clusters of high-value locations across Tampa where new EV charging stations would likely be most effective. High-suitability areas were concentrated near major transportation corridors such as Interstate 275, Interstate 4, and key north–south arterials, reflecting the strong influence of accessibility in the weighted model. Many of these locations also corresponded with higher population densities and developed land cover types, especially in central Tampa, around the University area, and along busy commercial corridors. Medium suitability zones appeared in suburban neighborhoods where population density is moderate and road connectivity is still relatively strong, suggesting secondary opportunities for long-term network expansion.

The environmental constraint mask played a major role in shaping the final pattern of suitability. Large portions of the western and southern waterfront, as well as low-lying areas along rivers and creeks, were removed from consideration due to the

presence of flood zones and wetlands. This exclusion is important because it prevents investment in infrastructure that could be damaged by flooding or negatively affect sensitive ecosystems. The contrast between high suitability zones on stable, developed land and low suitability zones in environmentally constrained areas highlights the trade-offs planners must navigate when siting EV infrastructure in coastal cities like Tampa.

When the modeled suitability results were compared with the locations of existing EV charging stations, several service gaps became apparent. Downtown Tampa and the immediate surrounding neighborhoods currently host many charging stations, which aligns with their classification as highly suitable in the model. However, northern Tampa, portions of the eastern urban fringe, and some residential corridors in the south showed relatively few existing stations despite having strong suitability scores. These underserved areas represent practical targets for future station placement because they combine favorable land conditions and population demand with limited current access. The final map of existing and potential EV charging locations (Figure 1) summarizes these spatial relationships and provides a clear visual tool for planners.

Figure 1. Existing and Potential EV Charging Locations in Tampa, FL



Conclusion

This project demonstrates how GIS-based suitability modeling can support data-driven planning for EV infrastructure in Tampa. By combining information about road accessibility, population density, land-cover conditions, and environmental constraints, the model identifies locations that balance user demand with safety and environmental stewardship. The results highlight both areas that are already well served by existing charging stations and neighborhoods where new stations could significantly improve access. While additional factors such as economic cost, electric grid capacity, and detailed land-ownership information were beyond the scope of this project, they could be incorporated into future iterations of the model. Overall, the analysis provides a replicable framework that the City of Tampa and similar municipalities can use to guide the expansion of EV charging infrastructure in a sustainable and equitable way.

References

City of Tampa GIS Open Data. Various datasets, 2024.

Florida Geographic Data Library (FGDL). Land use and environmental layers.

FEMA National Flood Hazard Layer. Flood zone data.

U.S. Census Bureau, American Community Survey (2023). Population density estimates.

OpenStreetMap Contributors and Florida Department of Transportation. Road network data.

National Wetlands Inventory (NWI). Wetlands and surface water data for Tampa Bay region.